

Trailing versus Centered Windows

1. Trailing window vs. centered window in moving averages

- **Trailing window** → Average is computed using the current value and a fixed number of past values.
Example: If your window is 3 days, today's moving average uses today, yesterday, and the day before yesterday.
- Works for forecasting because we can't see the future.
 - **Centered window** → Average is computed using values before *and* after a given point.
Example: If the window is 3 days, the moving average for Wednesday uses Tuesday, Wednesday, and Thursday.
- Gives smoother, more accurate smoothing for *historical* data.
- Not usable for real-time forecasting because future values are unknown.
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2. How it's applied here

- Present values → **Trailing windows** (can't look ahead).
 - Past values (when adjusting historical noise) → **Centered windows** (we *can* look ahead in history).
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3. Main point

Centered windows help produce smoother and more accurate results for past data smoothing, while trailing windows are the only option for real-time forecasting.

If you want, I can draw you a **simple diagram** showing the difference between trailing and centered windows so it's visually clear.
